Mathematical Foundations of Machine Learning Cheat Sheet

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Matrices and Vectors

Vector Addition: $x = \begin{bmatrix}0\\2\end{bmatrix}$ $y = \begin{bmatrix}2\\2\end{bmatrix}$ $x+y = \begin{bmatrix}2\\4\end{bmatrix}$

Scalar Multiplication: if $x = \begin{bmatrix}2\\2\end{bmatrix}$ then $2x = \begin{bmatrix}4\\4\end{bmatrix}$

Dot Product: $\hat{v} = \begin{bmatrix}3\\4\end{bmatrix}$ $\hat{w} = \begin{bmatrix}4\\3\end{bmatrix}$ $v \cdot w = 3\times4+4\times3 = 24$

- Distributive $u^T(v+w) = u^Tv + u^Tw$
- Commutative: $u^Tv = v^Tu$
- Not Associative: $u^T(v^Tw) \neq (u^Tv)^Tw$
- Vector Length/Magnitude: $\|v\| = \sqrt{3^2+4^2} = 5$

Unit Vectors: $\mu = \frac{1}{\|v\|}$

Matrix Addition: $A+B = \begin{bmatrix}a_11+b_11 & a_12+b_12\\ a_21+b_21 & a_22+b_22\end{bmatrix}$

Matrix Scalar Multiplication $\delta A = \begin{bmatrix}\delta a_{11} & \delta a_{12}\\ \delta a_{21} & \delta a_{21}\end{bmatrix}$

Transpose: $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ $A^T = \begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$

Matrix Trace: $A = \begin{bmatrix}\mathbf{3} & 2 & 4\\ 2 & \mathbf{5} & 6\\ 4 & 6 & \mathbf{8}\end{bmatrix}$ trace$(A) = 16$

Matrix Multiplication:

$A = \begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 1\end{bmatrix}, B = \begin{bmatrix}0 & 2\\ 1 & 1\\ 0 & 1\end{bmatrix}, AB = \begin{bmatrix}2 & 7\\ 2 & 9\end{bmatrix}$

Properties of Determinants

- det(I) = 1
- det(A) = 0 when two rows are equal
- det(A) = 0 when matrix has row of 0's
- if A is singular det(A) = 0, if A is invertible det(A) $\neq$ 0
- det(AB) = det(A)det(B)
- $det(A^T) = det(A)$

The Determinant

Example: $A = \begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 2\\ 0 & 0 & 1\end{bmatrix}$

$det(A) = (-1)^{1+1} \cdot 1\begin{vmatrix}1 & 2\\ 0 & 1\end{vmatrix} + (-1)^{1+2} \cdot 2\begin{vmatrix}3 & 2\\ 0 & 1\end{vmatrix}$

$+(-1)^{1+3} \cdot 3\begin{vmatrix}3 & 1\\ 0 & 0\end{vmatrix} = 1(1-0)-2(3-0+3(0-0) = -5$

The Inverse Matrix

Invertibility Test

- Must have non-zero pivots after elimination
- the det(A) must be non-zero
- Ax = 0, where x = 0 must be the only solution.

Gauss-Jordan Elimination $[A\|I] => [I\|A]$

e.g. for $A = \begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}$

$A^{-1} = \frac{1}{a_{11}a_{22}-a_{12}a21}\begin{bmatrix}a_{22} & -a_{12}\\ -a_{21} & a_{11}\end{bmatrix}$

$= \frac{1}{det(A)}\begin{bmatrix}a_{22} & -a_{12}\\ -a_{21} & a_{11}\end{bmatrix}$

e.g. for $A = \begin{bmatrix}1 & 2 & 1\\ 4 & 4 & 5\\ 6 & 7 & 7\end{bmatrix}$ Minors $= \begin{bmatrix}-7 & -2 & +4\\ +7 & +1 & -5\\ +6 & +1 & -4\end{bmatrix}$

Cofactors $= \begin{bmatrix}-7 & +2 & +4\\ -7 & +1 & +5\\ +6 & -1 & -4\end{bmatrix}$

Adjugate $= Cofactors^T = \begin{bmatrix}-7 & -7 & +6\\ +2 & +1 & -1\\ +4 & +5 & -4\end{bmatrix}$

$\mathrm{A}^{-1} = \begin{bmatrix}-7 & -7 & 6\\ 2 & 1 & -1\\ 4 & 5 & -4\end{bmatrix}$

Spaces/Subspaces

The four fundamental subspaces of A_{mxn}

Column Space	$C(A) \in \mathbb{R}^m$	$Ax = b$
Null Space	$N(A) \in \mathbb{R}^n$	$Ax = 0$
Row Space	$C(A^T) \in \mathbb{R}^n$	$A^Tx = b$
Left Null Space	$N(A^T) \in \mathbb{R}^m$	$A^Tx = 0$

The complete solution to Ax = b

- set all free variables to 0, solve Ax=b for pivot variables to find $X_{particular}$.
- find the null space: $X_{nullspace}$
- the complete solution is $x = x_{particular} + x_{nullspace}$

To solve for matrix rank, Guassian Elimination until REF, count the pivots. Linear Independence: z cannot be expressed as a linear combination of x and y.

Projections

The projection of b onto a line
$\mathbf{p} = \frac{a^Tb}{a^Ta}a$ $\mathbf{P} = \frac{aa^T}{a^Ta}$
The projection of $\mathbf{b}$ onto the subspace spanned by the columns of A
$\mathbf{p} = A(A^TA)^{-1}A^T\mathbf{b}$ $\mathbf{P} = A(A^TA)^{-1}A^T$
Two vectors are orthogonal when their dot product is 0.
If A has linearly independent columns then A^TA is invertible

Gram-Schmidt Process

- Let A=a
- $B = b - \frac{A^Tb}{A^TA}A$
- $C = c - \frac{A^Tc}{A^TA}A - \frac{B^Tc}{B^TB}B$
- $q_1 = \frac{A}{\|A\|} q_2 = \frac{B}{\|B\|} q_3 = \frac{C}{\|C\|}$

Matrix Decomposition

Finding Eigenvalues $det(A - \lambda I) = 0$

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ solve for $det(\begin{bmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix})$

$(1-\lambda)(2-\lambda) = 0$

$\lambda = 1, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = 0,\ v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda = 2, \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ 0 \end{bmatrix} = 0,\ v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Compute the SVD

A is given, $A = U\Sigma V^T$

$det(A^T A - \lambda I) = 0$

V is the matrix composed of v_1 and v_2 from calculating the determinant

$\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{bmatrix}$

U is found by calculating $u_1 = \frac{Av_1}{\delta_1}$, $\delta_1 = \frac{1}{\sqrt{\lambda_1}}$ and $u_2 = \frac{Av_2}{\delta_2}$, $\delta_2 = \frac{1}{\sqrt{\lambda_2}}$